

VIKRAM NAG ASHOKA

Robotics Data & ML Infrastructure | Real-Time Systems | Agentic AI
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SUMMARY

Staff engineer with 10+ years building the data backbone for large-scale intelligent systems: real-time pipelines processing 2.5B+ events/day, ML experimentation platforms, and multi-agent LLM systems. Now applying that depth to robotics — shipped open-source platforms for ROS log analysis and robot fleet observability.

SKILLS

- **Robotics & Telemetry:** ROS 2 bag tooling (roscpp, rosbag2, mcap), time-series robot log analysis, sensor data pipelines
- **Real-Time Data:** Kafka, Spark, Airflow, streaming enrichment & routing, sub-second analytics, Elasticsearch
- **Languages:** Python, C++, Scala, SQL
- **Agentic AI:** Multi-agent orchestration, LLM pipelines, browser/tool-use agents, evaluation harnesses
- **Cloud & Infra:** GCP (Dataproc, GCS, BigQuery), AWS, Terraform, Kubernetes, Docker, observability & alerting frameworks

OPEN-SOURCE ROBOTICS PROJECTS

RosBag Resurrector — *Open-source analysis & ML-export platform for ROS 2 sensor logs* 2026
Python · Polars · DuckDB · FastAPI · React/TypeScript · MCAP | 15 PyPI releases · 800+ automated tests | [GitHub](#) · [Demo video](#)

- Designed a bounded-memory streaming core that analyzes multi-GB robot sensor logs with memory bounded by chunk size, not bag size — an invariant enforced by a dedicated memory-regression CI suite.
- Built streaming export pipelines from raw MCAP into 7 analytics/ML formats (Parquet, HDF5, Zarr, LeRobot, RLDS) with multi-stream time synchronization.
- Shipped a data-quality layer — streaming health checks, declarative “bag contracts” usable as CI gates, one-command incident reports — answering “is this run clean enough to train on?”

Ground Control — *Robot fleet observability platform with counterfactual replay* 2026
Python · MuJoCo · WebSockets · Postgres/SQLite · Fly.io | 539 tests · 7-job CI matrix · deployed live | [Live app](#) · [Demo video](#)

- Built counterfactual replay for robot fleets — a capability no existing platform (Foxglove, Formant, Viam, InOrbit) offers: select any logged decision, change a command parameter, and re-run it through closed-loop MuJoCo physics with baseline-vs-counterfactual comparison.
- Engineered the streaming data path end-to-end: authenticated WebSocket ingest with per-frame ACKs, content-addressed dedup, a dual-engine SQLite/Postgres store, and 250 ms coalesced live fan-out with cursor resume.
- Designed a typed provenance model where the system refuses to render claims it can't back — physics-validated vs. policy-only results are distinguished end-to-end; the retraining gate correctly blocked its own 1M-step candidate model.

EXPERIENCE

Walmart Global Tech — *Staff Software Engineer* 2021 – Present
Staff Software Engineer (Nov 2024 – Present) · Senior Software Engineer (Oct 2022 – Nov 2024) · Software Engineer III (Jun 2021 – Oct 2022)

- Designed a Kafka + Spring Boot streaming framework processing ~2.5 billion events daily — dynamic routing of heterogeneous event types, in-flight enrichment, and persistence to GCS for real-time, sub-second analytics.
- Tech lead for Target-ROAS, an ML-driven optimization platform — led a team of 4 engineers, owned design of all data components, and drove POCs and core development; the system autonomously optimizes advertiser spend via A/B-tested ML pipelines and real-time experimentation.
- Built components of a multi-agent LLM system for conversational analytics — autonomous agents for query generation, data retrieval, insight synthesis, and summarization, orchestrated across reasoning and data layers.
- Partnered with data science and product on rapid prototyping, cutting algorithm iteration from weeks to days.

Moody's Analytics — *Software Engineer* 2018 – 2021

- Built Spark ETL pipelines, migrated legacy SQL workloads to Spark + Elasticsearch, and built reusable data modules for reproducible research.

Alcatel-Lucent (HCL) — *Software Engineer* — *BTS measurement tooling; RNC simulator (NBAP)* 2014 – 2016

EDUCATION, CERTIFICATIONS & PATENT

- **M.S. Computer Science** — University of Pennsylvania (2016 – 2018)
- **Udacity Nanodegree** — **Self-Driving Car Engineering & AI** (2017 – 2018): perception, sensor fusion, control
- **B.E. Computer Science** — BMS College of Engineering (2010 – 2014)
- **Patent:** [US-20250156899-A1](#) — Multichannel In-Club Attribution System for Featured Items